

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer at an early-stage startup, building backend product from the ground up: a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives inside the EHR prescribing workflow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that automates medication-alternative generation from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, sentence-transformers, React), owning architecture and all technical decisions end to end while productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** running hermetically in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research agent end to end in **TypeScript** that fetches web sources and streams multi-step digests over WebSocket, with durable workflows (WorkflowEntrypoint + Durable Objects) that survive process restarts and tool failures and persist state in D1 with no external database.

align-ops – LLM Post-Training + Serving Ops PyTorch, TRL, vLLM, FastAPI, Docker

- Built a backend serving stack, a **FastAPI** gateway over a vLLM server with Redis caching and async request micro-batching, lifting throughput **1.9x** and GPU utilization **52%→84%** at **640 ms** p95 time-to-first-token; containerized with Docker and instrumented with Prometheus/Grafana.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, Java, C++, SQL, Bash, MongoDB, PostgreSQL, SQLite
Backend & APIs: FastAPI, Node.js, Express, React, REST APIs, async services / queues, authentication & rate limiting
ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, LLM agents, RAG, evaluation
Systems & Testing: Spark / PySpark, Linux profiling, distributed systems, pytest, integration testing, CI/CD
Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Git, GitHub Actions

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.